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# Week 6, Lecture 3 - From Theory to Practice

**PSYC 51.07: Models of Language and Communication**

Winter 2026

# Today's Agenda 📋

- 🚀 **Real-World Applications**: Where BERT shines
- 🧠 **Cognitive Neuroscience**: Brain-model parallels
- 🤔 **Understanding vs. Pattern Matching**: The big debate
- ⚠️ **Limitations**: What BERT can't do
- 💡 **Practical Tips**: Deployment and optimization
- 🔮 **Future Directions**: Where are we heading?

*Goal: Connect BERT to real applications and understand broader implications*

# BERT Applications 🚀

**BERT excels at understanding tasks:**

**Classification Tasks:**

- Sentiment Analysis
- Topic Classification
- Spam Detection
- Intent Recognition

**Token-Level Tasks:**

- Named Entity Recognition (NER)
- Part-of-Speech Tagging
- Word Sense Disambiguation

**Span-Level Tasks:**

- Question Answering
- Extractive Summarization
- Information Extraction

**Sentence-Pair Tasks:**

- Semantic Similarity
- Natural Language Inference
- Paraphrase Detection

## Industry Impact

BERT powers:

- Google Search (understanding queries)
- Customer service chatbots
- Content moderation
- Document understanding

# Case Study: Google Search 🔍

**BERT revolutionized search in 2019**

**Problem:**

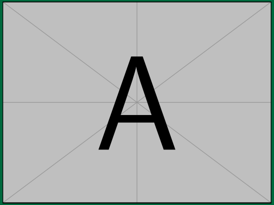
- Search queries often depend on context and word order
- Traditional keyword matching misses nuance
- Example: "2019 brazil traveler to usa need a visa"

**Before BERT:**

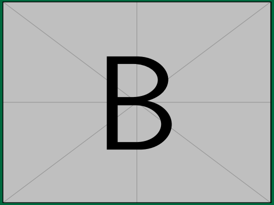
- Focused on "brazil" and "usa"
- Missed importance of "to"
- Returned results about US travelers to Brazil
- Wrong direction!

**With BERT:**

- Understands "to" is critical
- Grasps full context and direction
- Returns correct results about Brazilian travelers to US
- Correct understanding!




Wrong: US ▯ Brazil




Correct: Brazil ▯ US

*Google reported BERT improved 1 in 10 searches in English*

# Question Answering with BERT 💬

## Extractive QA: Find answer span in passage

**Example**

**Context:** ”The Normans (Norman: Nourmands; French: Normands; Latin: Normanni) were the people who in the 10th and 11th centuries gave their name to Normandy, a region in France.”

**Question:** ”In what country is Normandy located?”

**Answer:** France

```
from transformers import pipeline

# Load QA pipeline with BERT
qa_pipeline = pipeline("question-answering", model="bert-large-
    uncased-whole-word-masking-finetuned-squad")
```

# Named Entity Recognition 🏷️

## Token-level classification task

**Example**

**Input:** "Apple Inc. is headquartered in Cupertino, California."

**Output:**

- Apple Inc. ▯ ORGANIZATION
- Cupertino ▯ LOCATION
- California ▯ LOCATION

```
from transformers import pipeline

# Load NER pipeline
ner_pipeline = pipeline("ner", model="dslim/bert-base-NER")

# Extract entities
```

# Sentiment Analysis 😊😐😢

## Sequence classification task

**Examples**

- ”This movie was absolutely amazing!” ▯ POSITIVE
- ”The product broke after one week.” ▯ NEGATIVE
- ”The weather is cloudy today.” ▯ NEUTRAL

```python
from transformers import pipeline

# Load sentiment analysis pipeline
sentiment_pipeline = pipeline("sentiment-analysis", model="
    distilbert-base-uncased-finetuned-sst-2-english")

# Analyze sentiments
texts = [
```

# Semantic Similarity 🔗

## Measuring sentence similarity with BERT embeddings

```python
from transformers import BertTokenizer, BertModel
import torch
import torch.nn.functional as F

tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
model = BertModel.from_pretrained('bert-base-uncased')

def get_sentence_embedding(sentence):
    inputs = tokenizer(sentence, return_tensors='pt', padding=True
        , truncation=True)
    outputs = model(**inputs)
    # Use [CLS] token embedding as sentence representation
    return outputs.last_hidden_state[:, 0, :]

# Compare sentences
```

# Cognitive Neuroscience Perspective 🧠

**How do brains and models process language?**

**Predictive Processing in the Brain:**

- Brain constantly predicts upcoming input
- N400: Neural response to unexpected words
- P600: Syntactic anomaly detection
- Context shapes predictions
- Prediction errors drive learning

**Key Brain Regions:**

- **Left IFG**: Syntax processing
- **Left STG/MTG**: Semantic processing
- **ATL**: Conceptual knowledge

**Predictive Processing in Models:**

- BERT: Predict masked words
- GPT: Predict next word
- Both use context to predict
- Surprise = high loss
- Gradient descent = learning

**Similarities:**

- Both hierarchical
- Both context-sensitive
- Both predictive
- Both learn from errors

*References: Kuperberg & Jaeger (2016), Willems et al. (2016), Hagoort & Indefrey (2014)*

# Prediction in Brains vs. Language Models 🧠🤖

## Parallels between neural and artificial systems

| Human Brain | Transformer Models |
|---|---|
| N400 component<br>(surprise at unexpected word)<br>(low probability prediction) | High cross-entropy loss |
| Hierarchical processing<br>(sounds → words → sentences)<br>(tokens → phrases → meaning) | Layer-by-layer representations |
| Context integration<br>(prior discourse, world knowledge)<br>(attend to relevant context) | Self-attention mechanism |

# Neural Encoding with Language Models 🔬

**Can we predict brain activity from language models?**

**Approach:**

- Show participants text while recording brain activity (fMRI/EEG)
- Extract representations from language model (e.g., BERT layer 6)
- Train regression model: Model embeddings ▯ Brain activity
- Test: Can we predict brain responses to new text?

**Findings:**

- **Better models ▯ Better brain prediction**
- BERT outperforms Word2Vec at predicting brain activity
- Different layers correlate with different brain regions
- Middle layers best for semantic areas
- Suggests shared representations?

**But...**

Correlation ▯ Causation! Models might capture statistical regularities that brains use, without using the same mechanisms.

*Reference: Willems et al. (2016) - "Prediction during natural language comprehension"*

# Discussion: What Does the Model ”Understand”? 🤔

## Does BERT understand language?

**Evidence FOR understanding:**

- Captures syntax and semantics
- Resolves ambiguity
- Handles long-range dependencies
- Generalizes to new examples
- Predicts brain activity
- Solves complex tasks

*”If it acts like it understands, maybe it does?”*

**Evidence AGAINST understanding:**

- No grounding in physical world
- No sensory experience
- No social context
- Brittle to adversarial examples
- No common sense reasoning
- Only pattern matching?

*”Understanding requires more than statistical patterns”*

**Key Questions**

- What is the difference between understanding and correlation?
- Can meaning exist without grounding?
- Is human understanding fundamentally different?

**Class Discussion: What do YOU think?**

# Adversarial Examples and Brittleness ⚠️

## BERT can be fooled easily

### Example: Sentiment Analysis

**Original:** ”This movie was great!”
**Prediction:** POSITIVE □

**Modified:** ”This movie was great! [SEP] terrible terrible terrible terrible”
**Prediction:** NEGATIVE □

Just adding repeated negative words after [SEP] flips the prediction!

### Other Brittleness Issues

- **Word substitutions**: Replace ”good” with synonym ”excellent” □ wrong prediction
- **Paraphrases**: Semantically equivalent but different words □ different predictions

# Limitations of Current Models ⚠️

**Despite impressive performance, transformers have limitations:**

- **Quadratic Complexity**
  - Self-attention scales as $O(n^2)$
  - Limited context windows (512-4096 tokens)
  - Cannot process very long documents efficiently
- **No True Understanding**
  - Pattern matching vs. comprehension
  - Lack of common sense
  - No world model
- **Data Efficiency**
  - Requires massive training data
  - Humans learn language with much less data
  - Not biologically plausible
- **Biases and Fairness**
  - Inherits biases from training data
  - Can amplify stereotypes
  - Ethical concerns

# Bias in Language Models ⚖️

**Models reflect and can amplify societal biases**

**Examples of Bias**

- **Gender bias:**
  - ”The doctor said [MASK] was running late” ▯ he (90%)
  - ”The nurse said [MASK] was running late” ▯ she (80%)
- **Occupational stereotypes:**
  - Associates certain jobs with certain genders
  - ”Programmer” ▯ male pronouns
  - ”Secretary” ▯ female pronouns
- **Racial bias:**
  - Different sentiment for names associated with different races
  - Can perpetuate harmful stereotypes

**Mitigation Strategies:**

Debiasing techniques during training
Balanced and diverse training data
Post-processing and filtering
Human oversight and auditing
Ongoing research area!

# Practical Tips for Working with Transformers 💡

**Start with Pre-trained Models**

- Don't train from scratch (too expensive!)
- Use HuggingFace Model Hub
- Choose appropriate model size

**Fine-tuning Best Practices**

- Use small learning rate (1e-5 to 5e-5)
- Add warmup steps
- Monitor for overfitting
- Freeze early layers if data is limited

**Computational Efficiency**

- Use mixed precision training (FP16)
- Gradient accumulation for larger batch sizes
- Consider DistilBERT for faster inference
- Use FlashAttention when available

**Evaluation**

- Use task-specific metrics
- Test on out-of-distribution data
- Check for biases
- Visualize attention for interpretability

# Deployment Considerations 🚀

**Moving from research to production**

**Model Selection**

- Balance quality vs. speed
- DistilBERT for low-latency applications
- BERT-Base for balanced performance
- BERT-Large when quality is critical

**Optimization**

- Quantization: FP32 □ INT8 (4x smaller, faster)
- Model pruning: Remove unnecessary weights
- Knowledge distillation: Compress to smaller model
- ONNX Runtime for optimized inference

**Infrastructure**

- GPU acceleration for batch processing
- CPU optimization for single requests
- Caching for repeated queries
- Load balancing for high traffic

**Monitoring**

- Track latency and throughput
- Monitor prediction quality
- Detect distribution drift
- A/B testing for improvements

# Future Directions

**Where is the field heading?**

**Longer Context**
- Efficient attention mechanisms (linear, sparse)
- Models with 100K+ token context
- Better long-document understanding

**Multimodal Models**
- Vision + Language (CLIP, DALL-E)
- Audio + Language (Whisper)
- Grounded understanding

**Better Pre-training**
- More efficient objectives
- Curriculum learning
- Continual learning

**Smaller, More Efficient Models**
- Better compression techniques
- Lottery ticket hypothesis
- Edge deployment

**Addressing Limitations**
- Debiasing and fairness
- Robustness and adversarial training
- Common sense reasoning
- Interpretability and explainability

# Encoder vs Decoder Models Revisited 🔄

## Different models for different tasks

| | Encoder (BERT) | Decoder (GPT) |
|---|---|---|
| **Architecture** | Bidirectional self-attention | Unidirectional (causal) attention |
| **Pre-training** | Masked Language Modeling | Next token prediction |
| **Best for**<br>• Classification<br>• NER, QA<br>• Similarity<br>• Text completion<br>• Dialogue<br>• Creative writing | Understanding tasks:<br><br><br>Generation tasks: | |

# Discussion Questions 💭

**Understanding vs. Pattern Matching:**

Where do you draw the line?
Is there a test for ”true” understanding?
Does it matter for applications?

**Brain-Model Parallels:**

How useful are these comparisons?
What can neuroscience learn from AI?
What can AI learn from neuroscience?

**Bias and Fairness:**

Who is responsible for addressing bias?
Can we ever have completely unbiased models?
How do we balance accuracy and fairness?

**Future of NLP:**

Will encoder models remain relevant?
Are decoder-only models the future?
What’s the next big breakthrough?

# Assignment 4: Context-Aware Models 📝

**Hands-on experience with transformers!**

**Tasks:**

- **Implement Attention Mechanism**
  - Build scaled dot-product attention from scratch
  - Visualize attention weights
- **Fine-tune BERT**
  - Load pre-trained BERT
  - Fine-tune on sentiment analysis
  - Compare to baseline models
- **Analyze Contextual Embeddings**
  - Extract embeddings for polysemous words
  - Visualize how context changes representations
  - Compare BERT vs Word2Vec
- **Explore Different Architectures**
  - Compare BERT (encoder) vs GPT (decoder)
  - Test on different tasks
  - Analyze strengths/weaknesses
- **Research Component**
  - Read one paper from references
  - Write brief summary & critical analysis

**Due:** Check course website for deadline

# Summary: Weeks 5-6 🎯

**What we learned:**

- **Evolution of Context**
  - Seq2Seq → Attention → Transformers
  - From bottleneck to full parallelization
- **Transformer Architecture**
  - Self-attention, multi-head attention
  - Positional encoding, layer norm, residuals
  - Encoder-only (BERT), Decoder-only (GPT), Both (T5)
- **BERT & Variants**
  - Masked Language Modeling
  - Pre-train then fine-tune paradigm
  - RoBERTa, ALBERT, DistilBERT, ELECTRA
- **Applications**
  - Classification, NER, QA, similarity
  - Real-world impact (Google Search, etc.)
- **Broader Implications**
  - Brain-model parallels
  - Understanding vs. pattern matching
  - Limitations and future directions

# Resources & Further Reading 📚

**Key Papers:**

Vaswani et al. (2017) - Attention Is All You Need
Devlin et al. (2019) - BERT: Pre-training of Deep Bidirectional Transformers
Liu et al. (2019) - RoBERTa
Sanh et al. (2019) - DistilBERT
Dao et al. (2022) - FlashAttention

**Cognitive Neuroscience:**

Hagoort & Indefrey (2014) - The neurobiology of language beyond single words
Kuperberg & Jaeger (2016) - What do we mean by prediction in language comprehension?
Willems et al. (2016) - Prediction during natural language comprehension

**Tutorials:**

HuggingFace Course: Chapters 1, 7
The Illustrated Transformer:
https://jalammar.github.io/illustrated-transformer/
BertViz: https://github.com/jessevig/bertviz

# Looking Forward in the Course 🔮

**Where do we go from here?**

**Upcoming Topics:**

- **Week 7:** Decoder models and text generation (GPT family)
- **Week 8:** Scaling laws and large language models
- **Week 9:** Prompting, in-context learning, and instruction tuning
- **Week 10:** Alignment, RLHF, and ethical considerations

**The Journey Continues:**

- From understanding (BERT) to generation (GPT)
- From supervised learning to few-shot learning
- From narrow tasks to general-purpose models
- From academic research to societal impact

**The transformer revolution continues! 🚀**

# Questions? 🙋‍♀️

## Discussion Time

**Topics to discuss:**

- BERT applications
- Understanding vs. pattern matching
- Cognitive neuroscience connections
- Limitations and future work
- Assignment 4 questions

Thank you! 🙏

See you in Week 7 for GPT and text generation!